Virtual brush

Virtual brush
1. Virtual brush
2. Preview of the effect
    2.1. The USB camera
    2.2. The CSI camera

MediaPipe is a cross-platform machine learning framework that can process key point detection such as hands, faces, and poses in videos and images in real time.

1. Virtual brush

MediaPipe virtual brush can track the state of the fingers through hand detection, and switch between different modes according to the finger state:

When the index finger and the middle finger of the right hand are held together, it is in the selection state, and a color palette pops up at the same time. When the two fingertips move to the position of the corresponding color, that color is selected (black is the eraser);

When the index finger and the middle finger are separated, it is in the drawing state, and you can draw freely on the canvas.

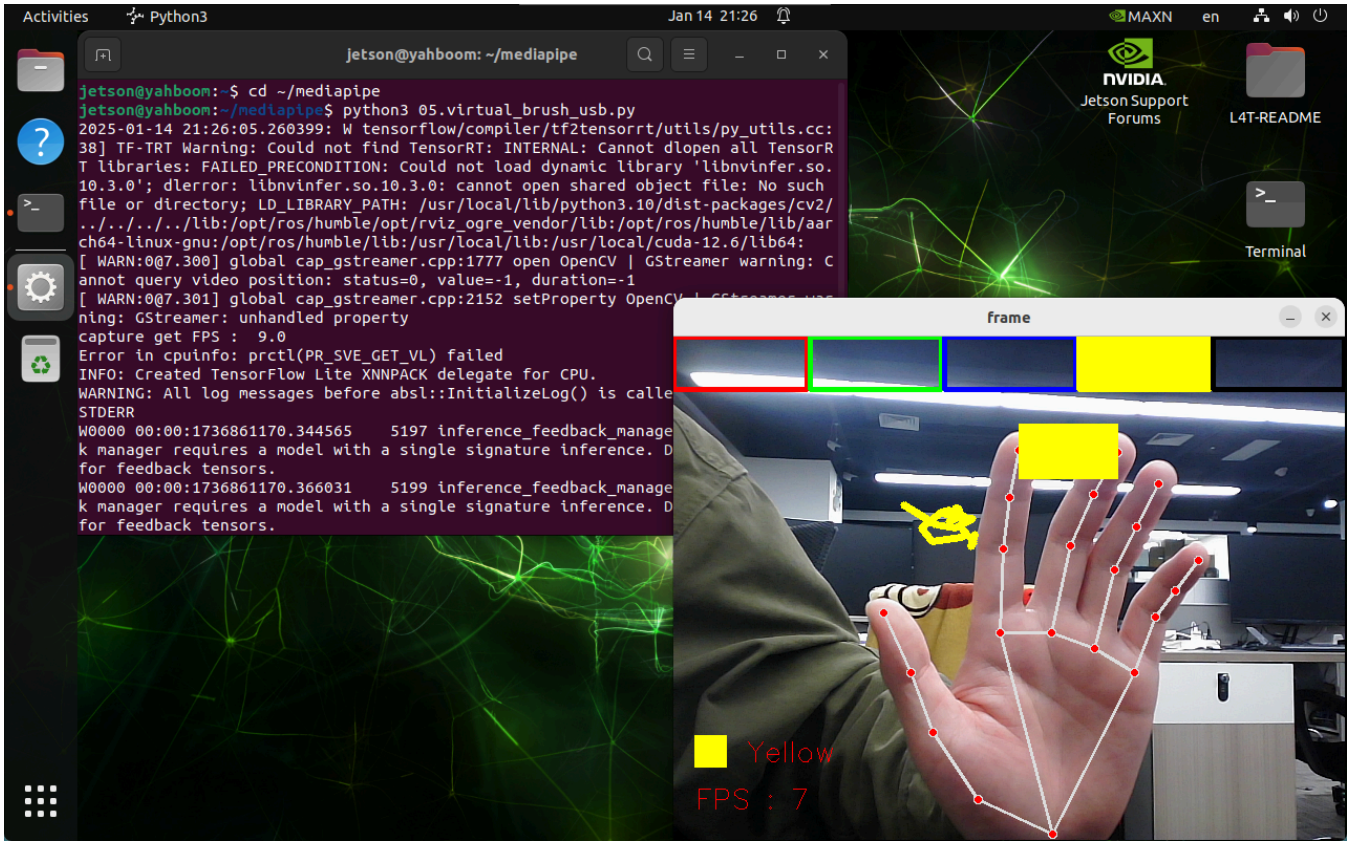
2. Preview of the effect

2.1. The USB camera

```
cd ~/mediapipe
```

```
python3 05.virtual_brush_usb.py
```

Note: click the camera window and press `q` to exit the program!

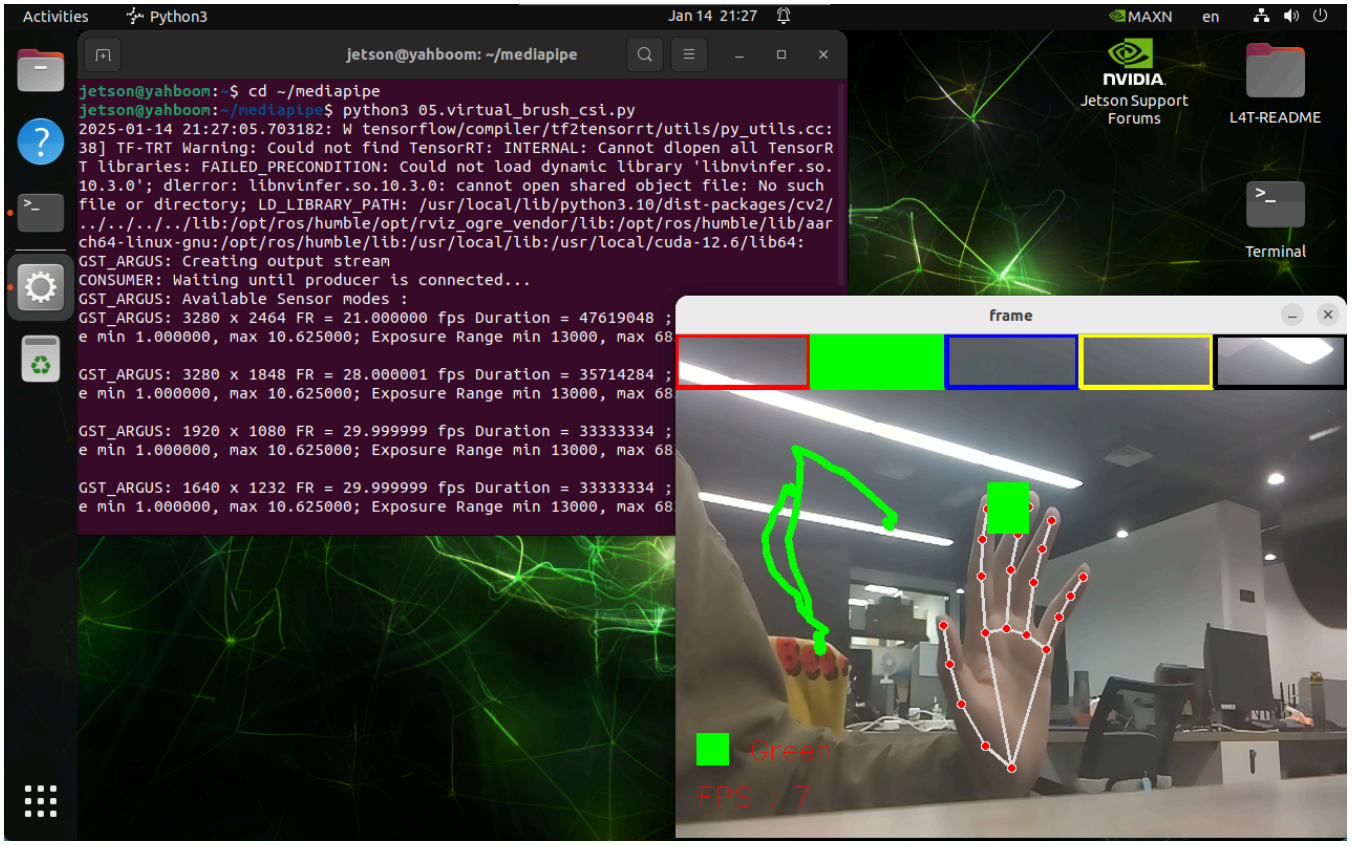


2.2. The CSI camera

```
cd ~/mediapipe
```

```
python3 05.virtual_brush_csi.py
```

Note: click the camera window and press `q` to exit the program!


Activities
Python3
Jan 14 21:27
MAXN
en
jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 05.virtual_brush_csi.py
2025-01-14 21:27:05.703182: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
NVIDIA
Jetson Support Forums
L4T-README
Terminal
frame
Green
FPS : 7